AI Playbook and Toolkit for Public Health Departments

Data Governance for AI-Ready Public Health Data

GOV 330

AI-ready data require governance before they require a model. Data governance defines ownership, stewardship, metadata, permissible use, data quality expectations, access rules, retention, lineage, and accountability. This module helps learners connect governance practices to safe AI use across surveillance, analytics, automation, public communication, and vendor-supported systems.

Level	intermediate/practitioner
Primary track	Governance and Security Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of data governance for ai-ready public health data in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI-ready data governance checklist that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

AI-ready data require governance before they require a model. Data governance defines ownership, stewardship, metadata, permissible use, data quality expectations, access rules, retention, lineage, and accountability. This module helps learners connect governance practices to safe AI use across surveillance, analytics, automation, public communication, and vendor-supported systems.

Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

Why This Topic Matters for Public Health AI

Data Governance for AI-Ready Public Health Data matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

Data governance is the set of decision rights, roles, policies, standards, and practices that guide how data are collected, managed, protected, shared, used, and retired. AI-ready data depend on governance because data must be understandable, authorized, high quality, and fit for use.

Data stewardship assigns responsibility for data quality, definitions, metadata, access, and appropriate use. Stewards do not necessarily own the data alone, but they help ensure that data are managed consistently and that users understand limitations.

Metadata describe the meaning, source, structure, quality, and permitted use of data. Without metadata, AI teams may misinterpret fields, misuse denominators, overlook missingness, or fail to recognize that a dataset is not appropriate for the intended use.

Public Health Example

A public health agency may want to use multiple datasets to prioritize outreach for vaccination. Data governance would clarify which datasets can be linked, which fields are authoritative, which populations are underrepresented, who can approve use, how long data can be retained, and how results can be shared. Without governance, the model may appear technical while resting on unclear authority and inconsistent data definitions.

Technical Deep Dive

AI-ready data begin with inventory. Agencies should know what datasets exist, who stewards them, what they contain, how often they are updated, what quality issues are known, and what uses are permitted. A data catalog can make this information visible to program, analytics, informatics, and governance teams.

Governance must also address data lineage and transformation. AI outputs may depend on mapped, linked, deduplicated, geocoded, imputed, or aggregated data. Staff need to understand how those transformations were performed and whether they are appropriate for the AI use case.

Permissible use should be explicit. Public health data may be collected under specific legal authorities or agreements. AI use may be appropriate for one purpose and inappropriate for another. Governance should define approved uses, prohibited uses, review pathways, and documentation requirements.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI-ready data governance checklist for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI-ready data governance checklist. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

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Knowledge Check

- 1. What is the best reason to address data governance for ai-ready public health data before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies